Applied Soft Computing Journal 163 (2024) 111875

Contents lists available at ScienceDirect

Applied Soft Computing

journal homepage: www.elsevier.com/locate/asoc

Co-guided Dual-channel Graph Neural Networks for the prediction of compound–protein interaction

Zheyu Wu [a], Huifang Ma [a,*], Bin Deng [a], Zhixin Li [b], Liang Chang [c]

[a] College of Computer Science and Engineering, Northwest Normal University, Lanzhou, Gansu 730070, China
[b] Key Lab of Education Blockchain and Intelligent Technology, Ministry of Education, Guangxi Normal University, Guilin, Guangxi 541004, China
[c] School of Computer Science and Information Security, Guilin University of Electronic Technology, Guilin, Guangxi 541004, China

ARTICLE INFO

Keywords:
Compound–Protein Interaction prediction
Graph Neural Network
Dual-channel schema
Co-guided learning
Drug discovery

ABSTRACT

Compound–Protein Interaction (CPI) serves as essential indicators for efficiently screening potential candidate drugs. Previous studies have typically focused on modeling CPIs either from intramolecular or intermolecular interactions, disregarding the diversity of interactions and the fine dependencies between these two types of interactions, thereby limiting the accuracy of CPI predictions. We argue that properly considering both intramolecular and intermolecular interactions allows for a more comprehensive understanding of the interactions between compounds and proteins. To this end, we propose a novel approach called Co-guided Dual-channel Graph Neural Network (CDGN) for CPI predictions. CDGN simultaneously captures various CPI information from intramolecular and intermolecular interactions using a dual-channel aggregating mechanism. Furthermore, to model the complicated relationships between the two interactions, we design a co-guided learning scheme to model the CPIs between intramolecular and intermolecular interactions, enhancing the learning of each other. Finally, we predict CPIs based on the rich interaction information from dual channels. Exhaustive experimental studies on two benchmarks verify the superiority of CDGN in CPI predictions. In particular, CDGN achieves outstanding performance with RMSE evaluation metrics of 1.263 and 1.626 on the publicly available PDBbind and CSAQ-HiQ datasets, respectively.

1. Introduction

Compound–Protein Interaction (CPI) refers to the interactions that occur between compounds and proteins in biological systems, which holds significant importance in drug discovery and development [1,2]. Compounds typically refer to drug candidates, which can include small molecules and biologics. These compounds possess the capacity to engage with proteins either as agonists or inhibitors, contributing to their potential role in biological processes for disease treatment [3]. Promising drug candidates are capable of interacting with proteins to form macromolecular complexes. Gaining a comprehensive understanding and accurate prediction of CPIs greatly facilitates the identification of potential drug targets and the optimization of drug efficacy [4,5]. Furthermore, this knowledge empowers researchers to gain insights into the mechanisms of drug action drugs and facilitates the design of novel therapeutic compounds with enhanced effectiveness and reduced side effects [6].

The interaction mechanisms between compounds and proteins are highly complicated, making the development of an efficient computational method capable of accurately determining CPIs a significant challenge. Initially, empirical methods [7–9] are developed with specialized designs for predicting docking and scoring functions, requiring domain expertise to encode internal biochemical interactions. Subsequently, statistical and machine learning-based techniques [10–14] emerge for predicting CPIs through data-driven learning. These methods aim to extract compound–protein features and utilize conventional regression models, such as random forest [10] and SVM [13]. However, these approaches heavily depend on hand-crafted features and lack generality when applied to larger datasets.

More recently, there has been a surge in the development of deep learning (DL) algorithms for predicting CPIs [15–28]. DL methods [17,18,24,25,27,28] generally employ Convolutional Neural Networks (CNNs) and pooling techniques to extract relevant patterns from the raw sequence information of both compounds and proteins. Among the various DL methods, Graph Neural Networks (GNNs) have demonstrated superior performance. In GNNs-based approaches [15,16,19–23,26], molecular graphs are used to naturally represent the structural information in molecules. Atoms (or sets of atoms like functional

* Corresponding author.
E-mail addresses: 2022222249@nwnu.edu.cn (Z. Wu), mahuifang@yeah.net (H. Ma), 2023212061@nwnu.edu.cn (B. Deng), lizx@gxnu.edu.cn (Z. Li), changl@guet.edu.cn (L. Chang).

https://doi.org/10.1016/j.asoc.2024.111875
Received 7 November 2023; Received in revised form 25 April 2024; Accepted 8 June 2024
Available online 12 June 2024
1568-4946/© 2024 Published by Elsevier B.V.

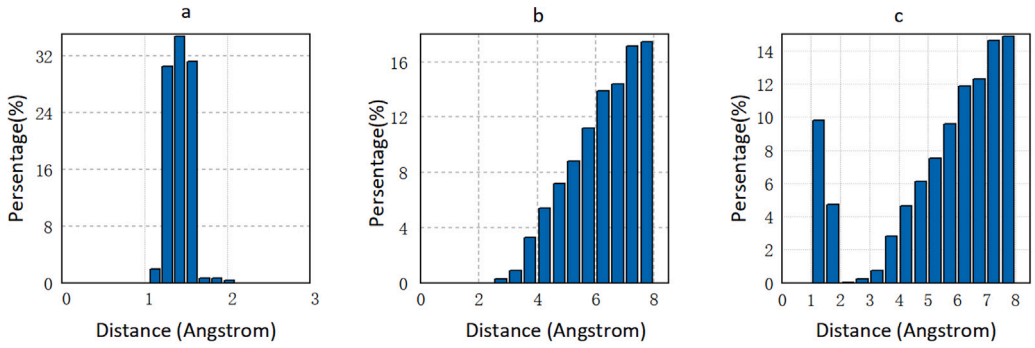


Fig. 1. The distribution of atom distance within an 8 Å radius in the compound–protein complex from the PDBbind core dataset. (a): The distribution of covalent bond edges. (b): The distribution of compound–protein atom pair edges. (c): The distribution of two kinds of edge distances.

groups) are treated as nodes, and the covalent bonds between atoms represent edges in the graph, reflecting the intramolecular interactions. GNNs can effectively learn the graph expression of compounds or proteins by aggregating topological information [16,19,20,23,26]. For example, GraphDTA [23] employs GNN models [29–31] on covalent edges to learn drug representations, combining them with protein representations to predict CPIs. Besides, some advancements have been made in GNN-based methods that incorporate spatial information to model macromolecule molecules, particularly for compound–protein complexes, as actual molecular interactions occur in three-dimensional (3D) space [15,21,22]. Typically, a cutoff interatomic distance, computed using atomic positions, is used to create noncovalent edges in molecular graphs to represent intermolecular interactions. For instance, Sign [21] captures intermolecular interactions by setting cutoff distances for noncovalent edges between different types of atoms.

Despite the success of GNNs methods, there are two main limitations that require further exploration. Firstly, existing research fails to adequately consider the diversity of interactions, including both covalent and noncovalent interactions. Covalent interactions (i.e., intramolecular interactions) occur within a single molecule and have a significant impact on its structure and properties. Noncovalent interactions (i.e., intermolecular interactions) involve forces between different molecules, including interactions between adjacent molecules. Therefore, incorporating both intramolecular and intermolecular interactions could be beneficial for predicting CPIs, and further investigation is needed in this regard. Secondly, the complex relations between intramolecular and intermolecular interactions have not been thoroughly investigated. Specifically, the strength and nature of intramolecular interactions can affect the occurrence and strength of intermolecular interactions. Similarly, the presence of intermolecular interactions can influence the nature and dynamics of intramolecular interactions. Therefore, integrating and understanding the mutual influence between these interactions is crucial for effective CPI predictions.

To motivate our work, we have conducted data analysis on the *core set* of PDBbind (details of data introduction is summarized in Section 4.1.2). We present the distribution of two types of edge distances: covalent bond edges and compound–protein atom pair edges, as shown in Fig. 1(a), (b), and (c), respectively. Specifically, we can observe that all interactions within or between molecules, such as covalent or non-covalent interactions, are distributed at different distances. The lengths of covalent bonds typically fall within the range of 1 to 2 Å (as illustrated in Fig. 1(a)), however, noncovalent interactions, such as hydrophobic interactions, van der Waals forces, and hydrogen bonds, are spread over longer distances (as shown in Fig. 1(b)). Therefore, these findings highlight that considering only single molecular interactions is insufficient for accurately predicting CPIs. Besides, we can also learn that the number of compound–protein atom pair edges are significantly different from covalent edges' (shown in Fig. 1(c)), so exploring the complicated relationships between the two kinds of edges is meaningful and domain related.

To alleviate both of the aforementioned limitations, this paper presents a novel model called **C**o-guided **D**ual-channel **G**raph Neural **N**etwork (CDGN) for CPI predictions. CDGN incorporates a dual-channel aggregating mechanism to simultaneously capture mutual reinforced information from both intramolecular and intermolecular interactions. In particular, for addressing the first challenge, the compound–protein complex is represented as three separate molecular graphs, each containing 3D structural and chemical data: the compound graph, the protein graph, and the bipartite compound–protein interaction graph. In the intra-type channel, two intramolecular graph convolution modules are utilized, one for the compound graph and the other for the protein graph. These modules extract intramolecular interactions from the chemical environment of connected atoms and bonds. The learning of intermolecular interaction representations is carried out by the intermolecular graph convolution module on the compound–protein interaction graph. For the second challenge, to capture the intimate relationship between these two types of interactions, a novel co-guided learning schema is proposed. Specifically, the intramolecular and intermolecular interactions learned from the dual-channel networks guide and enrich each other's semantics. The information from intramolecular interactions is used to adjust the weights of intermolecular interactions, while the information from intermolecular interactions modifies the energy and geometric configuration of intramolecular interactions. This mutual supplementation and collaboration between interactions within the compound and protein molecules, as well as between the two molecules, assists the model understand the interaction patterns of molecules furtherly. Finally, CDGN leverages the enhanced intramolecular and intermolecular interaction features to make CPI predictions.

In short, the dominating contributions of CDGN are as follows:

(1) We emphasize the critical significance of explicitly modeling intramolecular and intermolecular interactions synergistically to provide all-sided information of CPIs.

(2) We propose Co-guided Dual-channel Interactive Graph Neural Network (CDGN) to model intramolecular and intermolecular interactions and the relations between them to further boost the performance of CPI predictions.

(3) We conduct comprehensive experiments on two publicly available benchmark datasets to assess the performance of the proposed model, and prove that our CDGN has better accuracy and generalization ability.

The remainder of this paper is outlined as follows: Section 2 provides an introduction to fundamental concepts and presents a clear statement of the research problem. Section 3 offers detailed information about the overall pipeline and the components of our framework. In Section 4, we thoroughly analyze the experimental results obtained from our research. Finally, Section 5 presents the concluding remarks and provides a summary of our work.

Table 1
Key notations and explanations of CDGN framework.

Notation	Description
$\mathcal{G}$	The compound–protein complex graph
$\mathcal{G}_C$, $\mathcal{G}_P$, $\mathcal{G}_I$	The compound graph, the protein graph and the compound–protein interaction graph
$\mathcal{V}^C$, $\mathcal{V}^P$, $\mathcal{V}^I$	The atom node sets of $\mathcal{G}_C$, $\mathcal{G}_P$ and $\mathcal{G}_I$, respectively
$\mathcal{E}^C$, $\mathcal{E}^P$, $\mathcal{E}^I$	The edge sets of $\mathcal{G}_C$, $\mathcal{G}_P$ and $\mathcal{G}_I$, respectively
a_i^C, a_i^P, a_i^I	The ith atom node in $\mathcal{G}_C$, $\mathcal{G}_P$ and $\mathcal{G}_I$, respectively
e_{ij}^C, e_{ij}^P, e_{ij}^I	The directed edge from atom a_i to atom a_j in $\mathcal{G}_C$, $\mathcal{G}_P$ and $\mathcal{G}_I$, respectively
$\mathcal{M}^C$, $\mathcal{M}^P$	The 3D position matrices of compound and protein atoms in $\mathcal{G}$
$\mathbf{a}_i$, $\mathbf{e}_{ij}$	The embedding vectors of a_i^C and e_{ij}^C
$\mathbf{h}_{ra}(\mathbf{H}_{ra})$, $\mathbf{h}_{er}(\mathbf{H}_{er})$	The embedding vectors (enhanced) of intramolecular and intermolecular interactions
$\mathbf{g}_C$, $\mathbf{g}_P$	The embedding vectors of $\mathcal{G}_C$ and $\mathcal{G}_P$
$\mathbf{h}_i$, $\mathbf{h}_{ij}$	The embedding vectors of a_i^I and e_{ij}^P

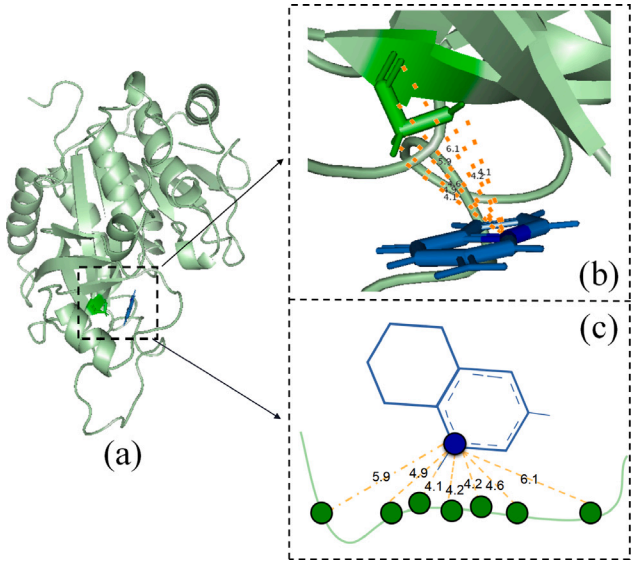


Fig. 2. An explanatory case transforming a original complex into a compound–protein interaction graph. (a): A 3D complex (PDB ID:3iny) graph from PDBbind dataset. (b): Interactions between an atom in the compound and atoms in a residue of the protein are visualized by yellow dashed lines with the distance between the two corresponding atoms. (c): Visualization of the atom-pair interactions by 2D graph to further explain (b).

2. Preliminaries

In this section, we present some notations used in CDGN and the process of the generation of three molecular graphs. In Table 1, a summary of the frequently used notations are introduced in this paper.

The Fig. 2(a) shows the 3D structure of the compound–protein complex $\mathcal{G}$, the atom node sets of compound and protein are defined as $\mathcal{V}^C = \{a^C_{1,\ldots,} a^C_m\}$ and $\mathcal{V}^P = \{a^P_{1,\ldots,} a^P_n\}$ with position matrix $\mathcal{M}^C \in \mathbb{R}^{m\times 3}$ and $\mathcal{M}^P \in \mathbb{R}^{n\times 3}$ for 3D coordinates of atoms, respectively. In this study, a total of three molecular graphs, specifically the compound graph, protein graph, and compound–protein interaction graph, are generated to represent the intramolecular and intermolecular interactions within a compound–protein complex.

Definition 2.1 (*Compound Graph*)**.** One compound molecule can be viewed as a graph with atoms as nodes and covalent bonds as edges. We define the compound graph as $\mathcal{G}_C = \langle \mathcal{V}^C, \mathcal{E}^C \rangle$ where $\mathcal{E}^C$ is the edge sets and the edge e_{ij}^C notes that the atom a_i^C and a_j^C are connected through a covalent bond in $\mathcal{G}$.

Definition 2.2 (*Protein Graph*)**.** Similar to the compound graph, we define the protein graph as $\mathcal{G}_P = \langle \mathcal{V}^P, \mathcal{E}^P \rangle$. Note that to ensure computational efficiency, only the protein atoms within the binding pocket are taken into account for graph construction.

Definition 2.3 (*Compound–Protein Interaction Graph*)**.** We define the compound–protein interaction graph shown in Fig. 2(b,c) as a bipartite graph $\mathcal{G}_I = \langle \mathcal{V}^I, \mathcal{E}^I \rangle$ and each edge connects an atom in the compound to one in the protein, where the vertex set is $\mathcal{V}^I \subseteq \mathcal{V}^C \cup \mathcal{V}^P$. Based on 3D positions of atoms, the interaction edge set $\mathcal{E}^I = f_i(\mathcal{V}^C, \mathcal{V}^P, \mathcal{M}^C, \mathcal{M}^P)$ are constructed claiming that the Euclidean distance between the compound atom and protein atom of an edge should be less than a definite cutoff threshold r_θ which is empirical and optimizable, and this definition supposes that specific interactions between pairs of atoms in the compound and protein exist within a defined threshold in $\mathcal{G}_I$.

Definition 2.4 (*Problem Statement*)**.** Given a compound–protein complex with 3D structure, i.e., the complex graph $\mathcal{G}$ and the 3D position matrix $\mathcal{M}$ consisting of $\mathcal{M}^C$ and $\mathcal{M}^P$, compound graph $\mathcal{G}_C$, protein graph $\mathcal{G}_P$ and compound–protein interaction graph $\mathcal{G}_I$ are created based on $\mathcal{G}$, and our objective is to train a model $f : (\mathcal{G}_C|\mathcal{G}_P|\mathcal{G}_I) \rightarrow y$ to accurately predict binding affinity y.

3. Model framework

3.1. Overview of the proposed CDGN

In this section, we present the comprehensive pipeline of our proposed CDGN framework, providing a detailed description of each component as illustrated in Fig. 3. We begin creating a compound graph, a protein graph and a compound–protein interaction graph from a compound–protein complex (Fig. 3(a)), which are preprocessed to corresponding molecular graphs with node and edge features. To effectively propagate information within and between these graphs, we then devise a dual-channel aggregating mechanism that operates in two channels: the intra-type channel (Fig. 3(b)) and the inter-type channel (Fig. 3(c)). This mechanism enables the extraction of embeddings for intramolecular interactions and intermolecular interactions, respectively. We also introduce a co-guided learning schema (Fig. 3(d)) to capture the complex relations between these two types of interaction embeddings. This schema facilitates the enhancement of each embedding through the cross transmission of interaction messages. Finally, we employ Multilayer Perceptron (MLP) layers to combine the two enhanced embeddings and make the final predictions for CPIs.

3.2. Dual-channel aggregating mechanism

To obtain intramolecular and intermolecular interactions, we utilize two channels, namely the intra-type channel and the inter-type channel, to aggregate messages among nodes and edges. In the intra-type channel, we employ two independent intramolecular graph convolution modules on the compound and protein molecular graphs to iteratively obtain node and edge embeddings. The edge representations from each intramolecular graph convolution layer are sequentially stacked to extract the final edge embeddings. To capture the graph embeddings, we apply global pooling on all edges in the compound and protein graphs separately. These two embeddings are then combined using

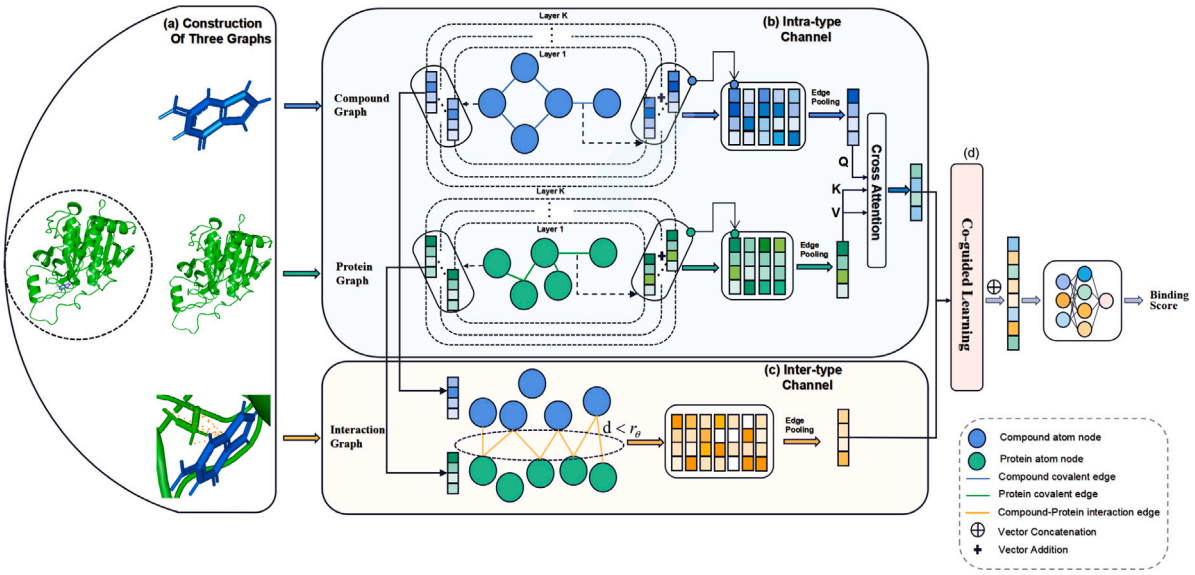


Fig. 3. The workflow of the proposed CDGN. (a): A simple illustration of the three graphs creation processes. (b): The process of intra-type channel. (c): The process of inter-type channel. (d): The illustration of co-guided learning schema.

a cross-attention module to model intramolecular interactions. In the inter-type channel, we apply an intermolecular graph convolution module to the bipartite compound–protein graph, inheriting initial atom states from the intramolecular graph convolution module to derive edge embeddings. Additionally, we include a global pooling layer for all edges to obtain a comprehensive representation of the bipartite compound–protein graph and capture intermolecular interactions.

3.2.1. Intra-type aggregating

This channel focuses on learning node and edge embeddings for $\mathcal{G}_C$ and $\mathcal{G}_P$, which are primarily influenced by non-covalent chemical environment. For $\mathcal{G}_C$ and $\mathcal{G}_P$, a set of 2D node features (such as atom type, atom degree, atom chirality, etc.) and 2D edge features (such as bond type, bond conjugation, bond ring, etc.) are encoded as initial features. Since CPIs are impacted by spatial distances and directions between atoms in three-dimensional space, we incorporate an additional set of 3D structural-related geometric features, encompassing distance, angle statistics, area statistics, and distance statistics, as supplementary edge features for both $\mathcal{GC}$ and $\mathcal{GP}$, which are encoded as one-hot vectors. To obtain the representation of the compound graph, we first perform a linear layer on initial $\mathbf{x}_i$ and $\mathbf{x}_{ij}$ of $\mathcal{G}_C$:

$$\mathbf{a}_i^0 = LeakyReLU(\mathbf{W}_a \cdot \mathbf{x}_i) \tag{1}$$

$$\mathbf{e}_{ij}^0 = LeakyReLU(\mathbf{W}_e \cdot [\mathbf{x}_i \parallel \mathbf{x}_{ij}]) \tag{2}$$

where $\mathbf{a}_i^0$ and $\mathbf{e}_{ij}^0$ are obtained by linear layer from the initial atom features with the dimension of d_a and bond ones of d_e. $\mathbf{W}_a \in \mathbb{R}^{D\times d_a}$ and $\mathbf{W}_e \in \mathbb{R}^{D\times(d_a+d_e)}$ are the transformation weight matrices, D is the dimension of the hidden state for the nodes and edges, the operator $\parallel$ represents concatenation. Then, following [32], we design an aggregation process that enables atom i to receive information from its neighboring edges using a weighted approach determined by the attention mechanism, which allows us to maximize the utilization of detailed information for interaction modeling:

$$\mathbf{m}_i^l = tanh\Big(\sum_{e_{ki}\in N_e(a_i)} z_{ki}^l \cdot \mathbf{W}_{e\to a}^l \cdot \mathbf{e}_{ki}^{(l-1)}\Big) \tag{3}$$

where $\mathbf{m}_i^l \in \mathbb{R}^{D\times D}$ effectively aggregates information about the neighboring edges of atom a_j at the $(l-1)$th layer, $\mathbf{W}_{e\to a}^l \in \mathbb{R}^{D\times D}$ is the learnable weight matrix and z_{ki}^l is the attention weight of the neighboring edge e_{ki} at the $(l-1)$th layer across the all incoming edges. Concretely, we first calculate the unnormalized coefficient s_{ij}^l between two connected nodes i and j and we apply softmax function to normalize the coefficient on neighboring edges of each node to get the attention weight at the $(l-1)$th layer:

$$s_{ij}^l = LeakyReLU(\mathbf{W}_{a_e}^l[\mathbf{a}_j^{(l-1)} \parallel \mathbf{e}_{ij}^{(l-1)}]) \tag{4}$$

$$z_{ij}^l = \frac{\exp(s_{ij}^l)}{\sum_{e_{ki}\in N_e(a_i)} \exp(s_{ki}^l)} \tag{5}$$

where $\mathbf{W}_{a_e}^l \in \mathbb{R}^{1\times 2D}$ is the transformation matrix for the combination of atoms and corresponding edges. In the subsequent phase of updating node hidden states, a GRU is used to combine the current incoming message with the previous hidden state, given its higher expressive capacity, to compute the new hidden state at the lth layer.

$$\mathbf{a}_i^l = BN(LeakyReLU(GRU(\mathbf{m}_i^l, \mathbf{a}_i^{(l-1)}))) \tag{6}$$

where GRU is the gated recurrent unit and BN is batch normalization operation with the aim of accelerating model training. After get the embeddings at the lth layer of all nodes, we add up the node embeddings of both sides of an edge and concatenate summation result with the embedding of e_{ij} at the $(l-1)$th layer to obtain the embeddings of $\mathbf{e}_{ij}$ at the lth layer:

$$\mathbf{e}_{ij}^l = MLP\Big(\Big[\mathbf{e}_{ij}^{(l-1)} \parallel (\mathbf{a}_i^l + \mathbf{a}_j^l)\Big]\Big) \tag{7}$$

where $\mathbf{a}_i^l$ and $\mathbf{a}_j^l$ are the node embeddings of both sides of e_{ij}.

Next, we iteratively update node and edge embeddings through k rounds of message passing to obtain the ultimate node and edge representations:

$$\mathbf{a}_i^{(*)} = \sum_{t=1}^{k} \mathbf{a}_i^k \tag{8}$$

$$\mathbf{e}_{ij}^{(*)} = \sum_{t=1}^{k} \mathbf{e}_{ij}^k \tag{9}$$

It is widely recognized that GCNs rely on the aggregation of neighboring information which inevitably suffer from oversmooth. This issue occurs when the embeddings of nodes and edges become increasingly similar as the aggregation radius (which is interpreted as the number of hidden layers in the GCNs) increases. The oversmooth problem reduces the diversity of nodes and edges and negatively impacts representation learning. To address this problem, we have introduced

a mixed representation aggregation technique with skip connections instead of relying solely on the embeddings from the last hidden layer for subsequent tasks. This method adds the embeddings $\mathbf{a}_i^k$ and $\mathbf{e}_{ij}^k$ in each hidden layer to produce the final embeddings $\mathbf{a}_i^{(*)} \in \mathbb{R}^D$ and $\mathbf{e}_{ij}^{(*)} \in \mathbb{R}^D$, respectively. This mixed aggregation method helps alleviate the oversmoothing issue as the aggregation radius increases, simultaneously enhancing the variety of nodes and edges. Furthermore, drawing inspiration from the fundamental physical principles governing compound–protein recognition, we exclusively employ the graph pooling module on the edges. These edges capture the acquired bonded interactions, which play a pivotal role in generating intramolecular interactions within complexes. Consequently, we employ a graph pooling function on final edge expressions to derive a comprehensive embedding for compound graph:

$$\mathbf{g}_C = Graph(\mathbf{E}) \tag{10}$$

$$Graph(\mathbf{X}) = [WeighedPool(\mathbf{X}) \parallel MaxPool(\mathbf{X})] \tag{11}$$

$$WeighedPool(\mathbf{X}) = Sigmoid(\mathbf{W}_g \cdot \mathbf{X}) \cdot \mathbf{X} \tag{12}$$

where $\mathbf{E} \in \mathbb{R}^{n_e^I \times D}$ is the feature map of compound edges, and the lth row of $\mathbf{E}$ is the final embedding of lth edge in $\mathcal{E}^C$. n_e^I is the number of edges of the compound graph, $\mathbf{g}_C \in \mathbb{R}^D$ is the compound graph embeddings, which is concatenated by two vectors generated from weighted pooling and max pooling, and $\mathbf{W}_g$ is parameters that can be learned. This method considers information from various perspectives, using max pooling to highlight edge features with the most significant information content and weighted sum pooling to utilize the overall information content of edge features.

Note: The compound graph convolution module and protein ones employ the same aggregation scheme for passing messages, as both are considered to represent covalent connections in their respective graphs. Consequently, the representation learning process for the protein graph $\mathcal{G}_P$ is identical to that of the compound graph $\mathcal{G}_C$. To avoid redundancy, we omit the specific procedures and only present the final representations $\mathbf{g}_P$ for the protein graph.

Once we have obtained the global embeddings for the compound graph and the protein graph separately, we employ a cross-attention module, as described in [33], to map the latent compound embeddings to the protein embeddings. This mapping process facilitates the construction of the final representation of intramolecular interactions. The resulting representations for the compound and protein, which capture their intramolecular interactions, can be calculated as follows:

$$\mathbf{Q} = f_Q(\mathbf{g}_C); \mathbf{K} = f_K(\mathbf{g}_P); \mathbf{V} = f_V(\mathbf{g}_P) \tag{13}$$

$$\mathbf{h}_{ra} = CrossAttention(\mathbf{Q}, \mathbf{K}, \mathbf{V}) \tag{14}$$

$$CrossAttention(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = Softmax(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{c/d}})\mathbf{V} \tag{15}$$

where $\mathbf{Q}$ is created from the compound graph embeddings, $\mathbf{Q}$ and $\mathbf{V}$ are generated from the protein graph embeddings, $\mathbf{h}_{ra}$ is the learned embeddings of the intramolecular interactions for the complexes, c and d are the embedding dimensions and number of heads, respectively. By employing the cross-attention mechanism, we can guide the model to capture patterns that demonstrate the influence of compound information on protein information. This allows the model to prioritize important intramolecular interaction information and thereby enhance the performance of the CPI prediction task.

3.2.2. Inter-type aggregating

Given that intermolecular interactions in $\mathcal{G}_I$ are predominantly driven by noncovalent/nonbonded interactions between compound atoms and protein atoms, we introduce a dedicated channel for learning edge representations. This channel effectively captures the pairwise atomic interactions between compound atoms and protein atoms, as outlined below:

$$\mathbf{h}_i = \mathbf{a}_i^{(*)}, \forall a_i \in (\mathcal{V}^C \cup \mathcal{V}^P) \tag{16}$$

$$\mathbf{h}_{ij} = MLP(\left[\mathbf{x}_{ij} \parallel (\mathbf{h}_i + \mathbf{h}_j)\right]), \forall e_{ij} \in \mathcal{E}^I \tag{17}$$

where the initial feature $\mathbf{h}_i \in \mathbb{R}^D$ of node i in $\mathcal{G}_I$ is generated from the corresponding last node representations in $\mathcal{G}_C$ and $\mathcal{G}_P$. The edge representation $\mathbf{h}_{ij} \in \mathbb{R}^{(D+1)}$ in $\mathcal{G}_I$ is calculated using Eq. (17). In this equation, the concatenation of initial atom features from two connected atoms and the initial edge feature is processed through MLP layers to produce the representations. $\mathbf{x}_{ij}$ represents the initial edge features, which encodes only the Euclidean distance between atom i and atom j, reflecting the physical generation of intermolecular interactions from pairwise atomic nonbonded interactions. Finally, we apply the same graph pooling function to the edge representation in order to obtain the global representation for the compound–protein interaction graph, capturing the intermolecular interactions as:

$$\mathbf{h}_{er} = Graph(\mathbf{h}_I) \tag{18}$$

where $\mathbf{h}_I \in \mathbb{R}^{n_e^I \times D}$ is the feature map of interaction edges, and the lth row of $\mathbf{h}_I$ is the embedding of lth edge in $\mathcal{E}^I$. n_e^I is the number of edges of the compound–protein interaction graph, $\mathbf{h}_{er} \in \mathbb{R}^D$ is the learned embeddings of the intermolecular interactions. In classical molecular force fields, the conversion from pairwise atomic interactions to overall nonbonded ones is typically achieved through theoretical calculation techniques. Nevertheless, in this work, we employ a neural network to implement this mapping, which automatically learns the concrete functional form that best represents the relationship between pairwise atomic interactions and the resulting total nonbonded interactions. This allows the model to capture complex and non-linear interactions that may not be adequately captured by traditional summation-based methods.

3.3. Co-guided learning schema

As previously mentioned, both intramolecular and intermolecular interactions have a mutual influence and jointly contribute to CPI predictions. Therefore, once we obtain the output embeddings for these two types of interactions, it is crucial to effectively integrate them to enable the model to capture informative embeddings that reveals CPI properties. Simple addition or concatenation is insufficient to handle this complex situation. Hence, we employ a co-guided learning schema to model the intricate relationships between the representations and enhance their mutual benefits. In this co-guided learning schema, we facilitate the cross transmission of information by cross-updating the weights associated with the two types of interactions. This mechanism enables the model to dynamically exchange and update information, enhancing the representation and understanding of the complex relations between intramolecular and intermolecular interactions. Inspired by [34], we first integrate the intramolecular interaction $\mathbf{h}_{ra}$ with the intermolecular interaction $\mathbf{h}_{er}$ in two ways:

$$\mathbf{m}_c = tanh(\mathbf{W}_1^{re}(\mathbf{h}_{ra} \cdot \mathbf{h}_{er}) + b_c) \tag{19}$$

$$\mathbf{m}_j = tanh(\mathbf{W}_2^{re}(\mathbf{h}_{ra} + \mathbf{h}_{er}) + b_j) \tag{20}$$

where $\mathbf{W}_1^{re}$, $\mathbf{W}_2^{re} \in \mathbb{R}^{D \times D}$, b_c and b_j are learnable parameters. The $\mathbf{m}_c$ and $\mathbf{m}_j$ signify interactive relations within distinct semantic spaces, representing intramolecular and intermolecular interactions. Subsequently, we apply a gating mechanism to furtherly capture the mutual dependencies between $\mathbf{h}_{ra}$ and $\mathbf{h}_{er}$ as follows:

$$\mathbf{r}_{ra} = LeakyReLU(\mathbf{W}_1^r \mathbf{m}_c + \mathbf{U}_1^r \mathbf{m}_j) \tag{21}$$

$$\mathbf{r}_{er} = LeakyReLU(\mathbf{W}_2^r \mathbf{m}_c + \mathbf{U}_2^r \mathbf{m}_j) \tag{22}$$

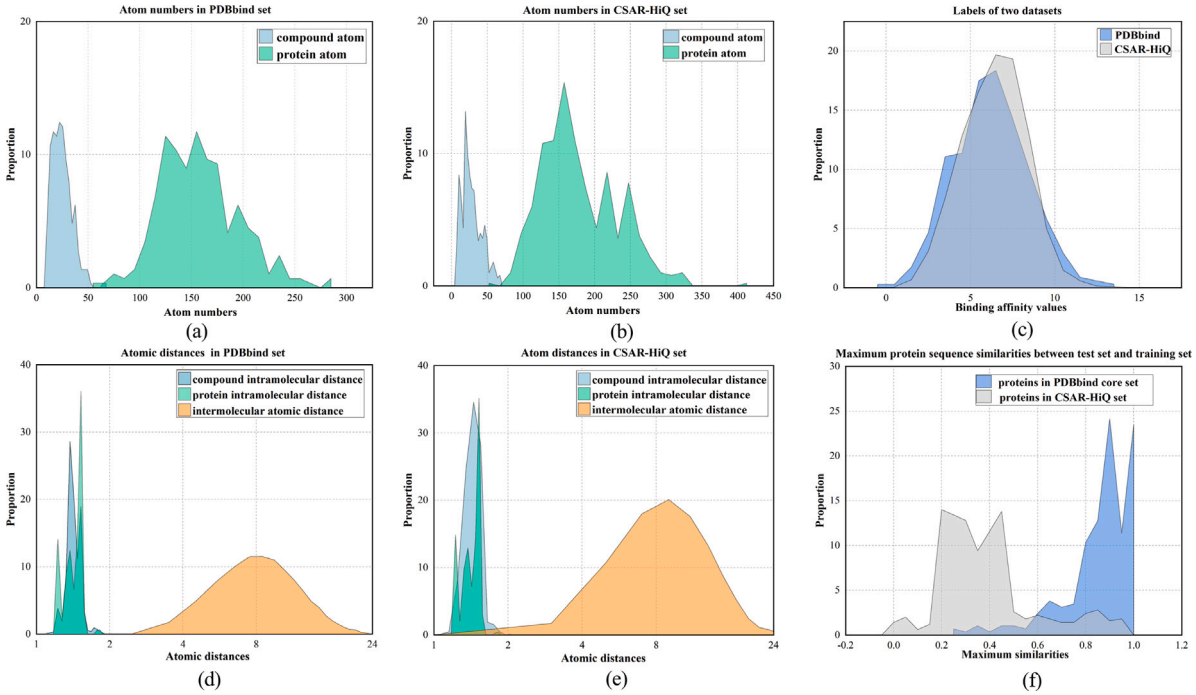


Fig. 4. Statistical distribution plots of key information for PDBbind set and CSAR-HiQ set. (a), (b): Distribution of the number of atoms of compounds and proteins in PDBbind set and CSAR-HiQ set, respectively. (c): Distribution of binding affinity label values of PDBbind set and CSAR-HiQ set, respectively. (d), (e): Distribution of intramolecular covalent bond lengths of compounds, intramolecular covalent bond lengths of proteins and intermolecular compound–protein atomic distances in PDBbind set and CSAR-HiQ set, respectively. (f): Distribution of maximum sequence similarities between proteins in the test sets (PDBbind *core set* and CSAR-HiQ set) and those in the training set (PDBbind *refined set*).

$$\mathbf{m}_{ra} = tanh(\mathbf{W}_{ra}(\mathbf{r}_{ra} \cdot \mathbf{h}_{ra}) + \mathbf{U}_{ra}((\mathbf{1} - \mathbf{r}_{er}) \cdot \mathbf{h}_{er})) \tag{23}$$

$$\mathbf{m}_{er} = tanh(\mathbf{W}_{er}(\mathbf{r}_{er} \cdot \mathbf{h}_{er}) + \mathbf{U}_{er}((\mathbf{1} - \mathbf{r}_{ra}) \cdot \mathbf{h}_{ra})) \tag{24}$$

where $\mathbf{W}_1^r$, $\mathbf{U}_1^r$, $\mathbf{W}_2^r$, $\mathbf{U}_2^r$, $\mathbf{W}_{ra}$, $\mathbf{U}_{ra}$, $\mathbf{W}_{er}$ and $\mathbf{U}_{er} \in \mathbb{R}^{D\times D}$. The $\mathbf{r}_{ra}$ ($\mathbf{r}_{er}$) denotes the remember gate which dictates the degree to which intramolecular (intermolecular) interactions are preserved while modeling their relationships. We also apply the complementary of $\mathbf{r}_{er}$ ($\mathbf{r}_{ra}$), i.e., $\mathbf{1} - \mathbf{r}_{er}$ ($\mathbf{1} - \mathbf{r}_{ra}$), to incorporate intramolecular (intermolecular) interactions to guide the learning of intramolecular (intermolecular) interactions, which enriches the semantic relevance of these two interactions. Finally, we get the enhanced intramolecular and intermolecular interactions as follows:

$$\mathbf{H}_{ra} = \mathbf{m}_{ra} \cdot (\mathbf{h}_{ra} + \mathbf{m}_{er}) \tag{25}$$

$$\mathbf{H}_{er} = \mathbf{m}_{er} \cdot (\mathbf{h}_{er} + \mathbf{m}_{ra}) \tag{26}$$

where $\mathbf{H}_{ra}$ and $\mathbf{H}_{er}$ represent the intramolecular and intermolecular interactions that mutually enrich each other in terms of their semantics. It is important to note that these intramolecular and intermolecular interactions ($\mathbf{H}_{ra}$ and $\mathbf{H}_{er}$) exchange information, guiding the learning process, which allows our model to effectively model the intricate relationships between them through two distinct channels for predicting compound–protein interactions.

3.4. Prediction and training

At the last part, two kinds of molecular interaction embeddings are concatenated to get final compound–protein interaction embedding and we use MLP to make final predictions:

$$\hat{y} = MLP(\mathbf{H}_{ra} \parallel \mathbf{H}_{er}) \tag{27}$$

Subsequently, in accordance with [24], we compute the loss by calculating the mean-square error (MSE) between the predicted $\hat{y}$ and the label value y for model optimization:

$$Loss = \frac{1}{n}\sum_{\mathcal{C}\in\mathcal{D}} (\hat{y} - y)^2 \tag{28}$$

where $\mathcal{D}$ includes all complexes with binding affinities and n is the number of the complexes in $\mathcal{D}$.

4. Experiments

4.1. Experiments setup

4.1.1. Research questions

This section is dedicated to performing experiments on two established datasets with the aim of addressing the following research inquiries:

- **RQ1** How does the performance of the CDGN model proposed in this study compare to that of the current state-of-the-art methods?
- **RQ2** Do all components in CDGN contribute to the performance?
- **RQ3** How is the interpretability of CDGN?
- **RQ4** What is the influence of the key hyper-parameters on CDGN?

4.1.2. Datasets and preprocessing

We evaluate all methods on two public standard datasets desired for CPIs:

- **PDBbind** is a famous public dataset [35] that provides 3D structural information of compound–protein complexes, along with experimentally determined binding affinities. In biological experiments, the binding affinity is typically related to dissociation constant (K_d), inhibition constant (K_i), or half inhibition concentration (IC_{50}) [36], and negative logarithm of these indexes are used on both the PDBbind and next CSAR-HiQ dataset. In our next experiment, we primarily use the PDBbind v2016 dataset, which is most frequently utilized in previous works [25,27]. Specifically, it contains three overlapping subsets, i.e., *general*, *refined* and *core set*. The *general set* includes 13,283

compound–protein complexes in total. However, the *refined set* consists of 4057 complexes selected from the *general set* due to their superior quality. In addition, the *core set* with 290 complexes serves as the highest quality benchmark for testing through a meticulous selection process [37]. For clarity, we name the contrast between the *refined* and *core* subsets, that is 3767 complexes, as *refined set* of PDBbind in the following.

• **CSAR-HiQ** is an additional benchmark dataset [38], incorporating two subsets, one containing 176 compound–protein complexes, and the other comprising 167, from an external dataset sourced independently. We utilize this external dataset to assess the model's overall generalization capabilities.

In order to uncover the distribution patterns of essential information within the two datasets and explore their similarities and differences, statistical analysis is conducted on the key information, as depicted in Fig. 4. Firstly, in both datasets, the distribution of the number of compound atoms in the complexes shows a concentration within small molecules consisting of up to 50 atoms. Conversely, the distribution of protein atoms spans a larger and wider range, as illustrated in Fig. 4(a) and (b). Secondly, the labels in both datasets exhibit similar Gaussian distributions (Fig. 4(c)). Additionally, we analyze the atomic distance distributions for different types of molecular interactions. The intramolecular interactions are mainly concentrated between 1 Å and 2 Å, whereas intermolecular interactions extend to longer distances (Fig. 4(d), (e)), posing a challenge in setting an appropriate cutoff distance to capture various CPI information. Lastly, by calculating the maximum sequence similarity [20] of target proteins in the complexes of the two test sets with those in the complexes of the training set, we find that the overlap between proteins in the CSAQ-HiQ set and those in the training set is much lower than that in the PDBbind *core set* (Fig. 4(f)). Therefore, the heterogeneous CSAQ-HiQ dataset can help assess the model's generalization ability.

To ensure fairness, we follow the common data partitioning method used in previous works [10,21,25,27,39]. On the whole, we select the refined set of PDBbind as stated in [10] for training set. To establish a training and validation set, we split the training set randomly within the refined set using a 9:1 ratio. For the testing sets, core set and CSAR-HiQ set are used, with the provision that we exclude any complexes already included in the refined set. This ensures that the testing sets consist of unique complexes that are not part of the training or validation process. Furthermore, recognizing that even lower-quality data from the general set can also contribute to enhancing model's performance [39], we perform an additional experiment using a larger but lower-quality general set for training to assess our model's generalizability. To create a validation set, we follow the approach outlined in [25,27] and randomly select 1000 complexes from the refined set. The remaining 11,993 complexes within the general set are utilized for training the model. This split allows us to evaluate the model's ability to generalize on previously unseen data.

4.1.3. Evaluation metrics

In order to thoroughly evaluate the performance of our model, we adopt the evaluation metrics commonly used in regression analysis, as suggested in [25,27,40]. These metrics include: Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Pearson's correlation coefficient (R), Standard Deviation (SD) and Kling–Gupta Efficiency (KGE). By utilizing these evaluation metrics, we can comprehensively assess the accuracy, precision, and correlation of our model's predictions with the true values in CPI predictions.

4.1.4. Baselines

To evaluate the performance of the proposed model CDGN, we choose following competitive methods as baselines:

• **ML-based Methods** encompass linear regression (LR), support vector regression (SVR) and random forest (RF), which adopt intermolecular interaction information mentioned in RF-Score [10] to make predictions between compounds and proteins.

• **CNN-based Methods:**

Pafnucy [25] is an illustrative 3D CNN-based model designed to capture the spatial arrangement within compound–protein complexes.

OnionNet [27] generates local and non-local interactions by grouping contact points based on the non-rotating free pair between ligand and protein atoms.

• **GraphDTA Methods:**

GraphDTA [23] incorporates GNN models to understand intricate graph structures and employs CNN to analyze protein sequences. It includes four variations, each utilizing three GNN methods: GCN [30], GAT [29], GIN [31], and a combination of GAT and GCN referred to as GAT-GCN.

• **GNN-based Methods:**

SGCN [41] makes use of node positions through a graph convolutional network, which directly incorporates atomic coordinates.

GNN-DTI [22] introduces a distance-aware graph attention algorithm for distinguishing between different types of intermolecular interactions by extracting the graph features of molecular interactions directly from the 3D structure of the protein–ligand binding site.

DimeNet [30] uses directional information related to a direction in coordinate space based on angular transformation of information by embedding information passed between atoms.

CMPNN [42] achieves downstream task improvements by designing a message transfer mechanism between edges and nodes to fully express molecular graphs.

SIGN [21] facilitates the 3D interaction and information exchange between edges and nodes, allowing for better prediction of binding affinity, particularly capturing long-range interactions.

4.1.5. Implementation details

CDGN model is implemented using the public DGL library (version: 1.0.2+cu116) with PyTorch as the backend (version: 1.13.1+cu116). The RDKit library (version: 2022.9.5) is used for handling chemical structures in Python. For training the CDGN model, we employ the Adam optimizer with a learning rate of 0.003. The batch size is set to 32. To prevent overfitting, we utilize early stopping during training. The training will be halted if performance in validation has no improvement for over 50 successive epochs. Based on following results obtained from the hyper-parameter study in Section 4.5, the cutoff threshold (r_θ) for the compound–protein interaction graph is set to 8 Å and the convolution layers (k) in the intramolecular module is set to 2. The hidden state dimension for nodes and edges in the model is set to 256, and a dropout rate of 0.1 is applied to mitigate overfitting.

4.2. Performance comparison (RQ1)

We begin by evaluating the performance of all methods on two standard benchmark datasets and subsequently investigate how the incorporation of diverse interactions enhances the model's ability to generalize on unseen and lower-quality data.

4.2.1. Overall comparison

From Table 2, CDGN demonstrates the best performances with superiority significances on both datasets among all methods. Specially, average significances of CDGN versus the best baseline models under five evaluation metrics on the PDBbind *core set* ($^{*}P = 2.7 \times 10^{-2}$) and CSAR-HiQ dataset ($^{**}P = 9.9 \times 10^{-4}$) are less than 0.05, which are considered to have significant differences. Moreover, we have made the following observations:

• From the perspective of capturing multi-type interaction information, GraphDTAs exhibit relatively poorer results indicating that intramolecular interaction only is deficient to model rich interactions between compound and protein, leading to shrinking performance. Besides, GraphDTAs fall short in modeling the complex structure of proteins by solely relying on protein sequence information. Models such

Table 2
The performance of CDGN and baselines on PDBbind *core set* and CSAR-HiQ set. We mark the best results in bold and the second-best results with underline. We report the mean and standard deviation of the results from five random runs. From two-sided t-test, we compute the average significances under five evaluation metrics of CDGN versus second-best baseline across two datasets. Two-sided t-test results: $^{*}P = 2.7 \times 10^{-2}$ for PDBbind *core set* and $^{**}P = 9.9 \times 10^{-4}$ for CSAR-HiQ set.

Method		PDBbind core set					CSAQ-HiQ set				
		RMSE ↓	MAE ↓	SD ↓	R ↑	KGE ↑	RMSE ↓	MAE ↓	SD ↓	R ↑	KGE ↑
ML-based methods	LR	1.675 (0.000)	1.358 (0.000)	1.612 (0.000)	0.671 (0.000)	0.588 (0.000)	2.071 (0.000)	1.622 (0.000)	1.973 (0.000)	0.652 (0.000)	0.573 (0.000)
	SVR	1.555 (0.000)	1.264 (0.000)	1.493 (0.000)	0.727 (0.000)	0.631 (0.000)	1.995 (0.000)	1.553 (0.000)	1.911 (0.000)	0.679 (0.000)	0.613 (0.000)
	RF-SCore	1.446 (0.008)	1.161 (0.007)	1.335 (0.010)	0.789 (0.003)	0.699 (0.013)	1.947 (0.012)	1.466 (0.009)	1.796 (0.020)	0.723 (0.007)	0.654 (0.028)
CNN-based methods	Pafnucy	1.585 (0.013)	1.284 (0.021)	1.563 (0.022)	0.695 (0.011)	0.632 (0.058)	1.939 (0.103)	1.562 (0.094)	1.885 (0.071)	0.686 (0.027)	0.623(0.093)
	OnionNet	1.407 (0.034)	1.078 (0.028)	1.391 (0.038)	0.768 (0.014)	0.734 (0.046)	1.927 (0.071)	1.471 (0.031)	1.877 (0.097)	0.690 (0.040)	0.685 (0.086)
GraphDTA methods	GCN	1.735 (0.034)	1.343 (0.037)	1.719 (0.027)	0.613 (0.016)	0.585 (0.041)	2.324 (0.079)	1.732 (0.065)	2.302 (0.061)	0.464 (0.047)	0.573 (0.081)
	GAT	1.765 (0.026)	1.354 (0.033)	1.740 (0.027)	0.601 (0.016)	0.581 (0.081)	2.213 (0.053)	1.651 (0.061)	2.215 (0.050)	0.524 (0.032)	0.568 (0.693)
	GIN	1.640 (0.044)	1.261 (0.044)	1.621 (0.036)	0.667 (0.018)	0.592 (0.066)	2.158 (0.074)	1.624 (0.058)	2.156 (0.088)	0.558 (0.047)	0.579 (0.720)
	GAT-GCN	1.562 (0.022)	1.191 (0.016)	1.558 (0.018)	0.697 (0.008)	0.636 (0.016)	1.980 (0.055)	1.493 (0.046)	1.969 (0.057)	0.653 (0.026)	0.615 (0.035)
GNN-based methods	SGCN	1.583 (0.033)	1.250 (0.036)	1.582 (0.320)	0.686 (0.015)	0.620 (0.050)	1.902 (0.063)	1.472 (0.067)	1.891 (0.077)	0.686 (0.030)	0.600 (0.053)
	GNN-DTI	1.492 (0.025)	1.192 (0.032)	1.471 (0.051)	0.736 (0.021)	0.667 (0.042)	1.972 (0.061)	1.547 (0.058)	1.834 (0.090)	0.709 (0.035)	0.632 (0.070)
	DimeNet	1.453 (0.027)	1.138 (0.026)	1.434 (0.023)	0.752 (0.010)	0.693 (0.050)	1.805 (0.036)	1.338 (0.026)	1.798 (0.027)	0.723 (0.010)	0.657 (0.067)
	CMPNN	1.408 (0.028)	1.117 (0.031)	1.399 (0.025)	0.765 (0.009)	0.723 (0.013)	1.839 (0.096)	1.411 (0.064)	1.767 (0.103)	0.730 (0.052)	0.682 (0.037)
	SIGN	1.316 (0.031)	1.027 (0.025)	1.312 (0.035)	0.797 (0.012)	0.752 (0.013)	1.735 (0.031)	1.327 (0.040)	1.709 (0.044)	0.754 (0.014)	0.715 (0.030)
Ours	CDGN	**1.263 (0.024)**	**1.005 (0.021)**	**1.228 (0.031)**	**0.849 (0.004)**	**0.796 (0.032)**	**1.626 (0.030)**	**1.245 (0.026)**	**1.540 (0.043)**	**0.802 (0.009)**	**0.745 (0.053)**

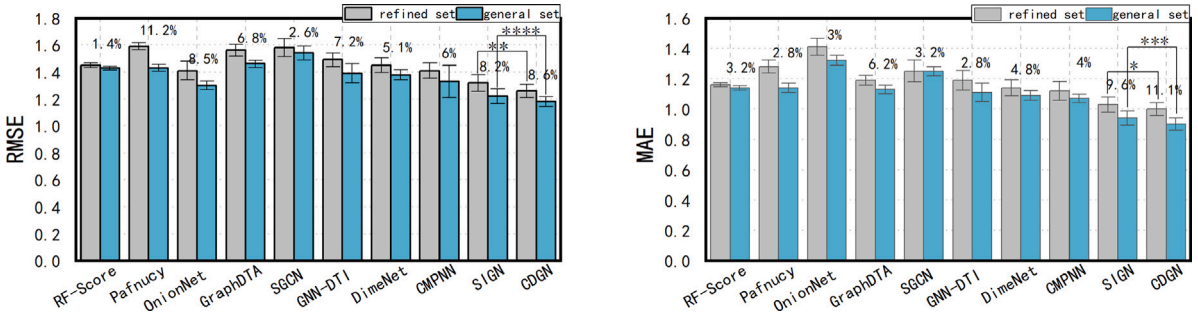
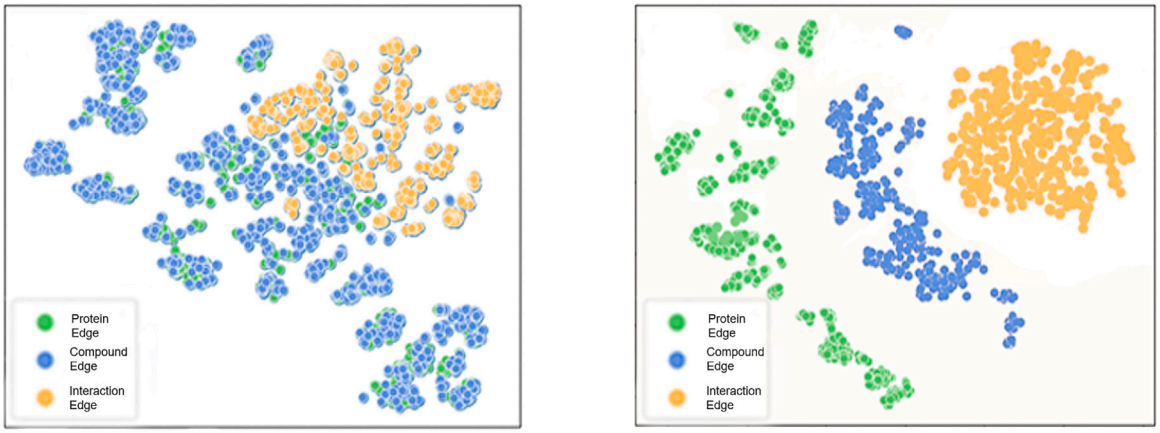


Fig. 5. Performance comparison among all methods on *core set* of PDBbind training on *general set*. We report the mean and standard deviation of the results from five random runs. The standard deviation is represented in the figure as error bars. The numbers between the two bars indicate the improvement rate relative to training on the *general set* compared to training on the *refined set*. We also show the performance significance of CDGN versus the second-best baseline (Two-sided t-test results: $^{****}P = 5.3 \times 10^{-5}$, $^{***}P = 4.5 \times 10^{-4}$, $^{**}P = 2.5 \times 10^{-3}$ and $^{*}P = 1.2 \times 10^{-2}$.).

Table 3
The ablation study results obtained on the PDBbind *core set* and CSAR-HiQ set. We mark the best results in bold. We report the mean and standard deviation of the results from five random runs. From Analysis of Variance, we calculate the average significances under five evaluation metrics among different model groups. Analysis of Variance results: $^{*}P = 1.0 \times 10^{-4}$ for (GraphDTA, Model-1 and Model-2 for PDBbind), $^{***}P = 1.6 \times 10^{-5}$ for (CDGN, Model-3, Model-4 and Model-5 for PDBbind), $^{**}P = 7.5 \times 10^{-5}$ for (GraphDTA, Model-1 and Model-2 for CSAR-HiQ) and $^{*****}P = 5.6 \times 10^{-6}$ for (CDGN, Model-3, Model-4 and Model-5 for CSAR-HiQ).

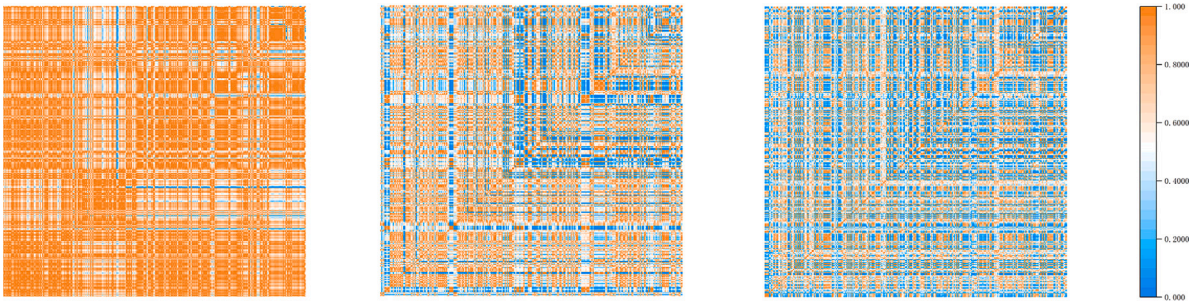
Dataset	Model	Intra-type channel	Inter-type channel	Co-guided schema	RMSE ↓	MAE ↓	SD ↓	R ↑	KGE ↑
PDBbind	GraphDTA	✓	–	–	1.735 (0.034)	1.343 (0.037)	1.719 (0.027)	0.613 (0.016)	0.585 (0.041)
	Model-1	✓	✓	–	1.682 (0.051)	1.270 (0.049)	1.654 (0.029)	0.637 (0.010)	0.612 (0.049)
	Model-2	✓	✓	✓	1.554 (0.032)	1.184 (0.025)	1.544 (0.049)	0.653 (0.003)	0.630 (0.056)
	CDGN	✓	✓	✓	**1.263 (0.024)**	**1.005 (0.021)**	**1.228 (0.031)**	**0.849 (0.004)**	**0.796 (0.032)**
	Model-3	–	✓	–	1.524 (0.045)	1.179 (0.033)	1.536 (0.039)	0.715 (0.017)	0.649 (0.068)
	Model-4	✓	–	–	1.493 (0.036)	1.168 (0.039)	1.489 (0.024)	0.689 (0.006)	0.666 (0.050)
	Model-5	✓	✓	–	1.332 (0.026)	1.110 (0.031)	1.326 (0.025)	0.795 (0.002)	0.729 (0.043)
CSAR-HiQ	GraphDTA	✓	–	–	2.324 (0.079)	1.732 (0.065)	2.302 (0.061)	0.464 (0.047)	0.573 (0.081)
	Model-1	✓	✓	–	2.138 (0.064)	1.616 (0.055)	2.148 (0.077)	0.556 (0.036)	0.589 (0.073)
	Model-2	✓	✓	✓	1.989 (0.065)	1.486 (0.053)	1.979 (0.055)	0.663 (0.031)	0.609 (0.062)
	CDGN	✓	✓	✓	**1.626 (0.030)**	**1.245 (0.026)**	**1.540 (0.043)**	**0.802 (0.009)**	**0.745 (0.053)**
	Model-3	–	✓	–	1.901 (0.065)	1.474 (0.067)	1.881 (0.078)	0.696 (0.029)	0.621 (0.074)
	Model-4	✓	–	–	1.808 (0.039)	1.338 (0.036)	1.782 (0.026)	0.729 (0.016)	0.630 (0.069)
	Model-5	✓	✓	–	1.705 (0.029)	1.316 (0.034)	1.696 (0.039)	0.768 (0.011)	0.706 (0.055)

as CMPNN, SGCN, and DimeNet only capture intramolecular interactions on a single-type molecular graph, employing different message passing schemes. GNN-DTI considers different types of intermolecular interactions through distance awareness. However, the limitations of these approaches stem from the incomplete information they rely on, which ultimately hampers the accuracy of CPI predictions. The underperformance of above baseline models can be attributed to their insufficient consideration of molecular interactions and their inability to comprehensively model the intricate structure of proteins and compounds in CPI prediction task.



(a) Before Training (b) After Training

Fig. 6. The clustering distribution visualization before and after training of the embedding vectors of the three types of edges of 50 complexes from PDBbind *core set* using t-distributed stochastic neighbor embedding (t-SNE).



(a) (b) (c)

Fig. 7. Plot of heat map for similarity matrix of 290 complexes in PDBbind *core set* under CDGN and its variants. (a): Without training. (b): Training without co-guided learning schema. (c): Training with CDGN.

• From the perspective of mining complex associations between interactions, machine learning-based methods such as OnionNet, Pafnucy, and the SIGN model leverage both intramolecular and intermolecular interaction features, resulting in improved performance. However, these baselines do not fully exploit the intricate and nuanced connections between intramolecular and intermolecular interactions resulting in poorer performance. As a result, their ability to generalize to new data, such as the CSAR-HiQ dataset, may be limited.

• Our proposed CDGN consistently yields the best performance on all the datasets. We attribute this to its unique architecture and design to effectively integrate and exploit the rich information from both intramolecular and intermolecular interactions. By doing so, CDGN can uncover and capture the intricate relationships and dependencies between these two types of interactions, leading to enhanced modeling capabilities. This comprehensive approach results in improved performance in modeling CPIs. In summary, CDGN surpasses existing methods by not only capturing a broader range of interaction information but also by exploring and modeling the complex interactions between intramolecular and intermolecular interactions.

4.2.2. Generalizability comparison

Due to the lower similarity between proteins in the heterogeneous CSAR-HiQ dataset and those in the training set (as shown in Fig. 4(f)), the model fails to adequately learn unseen target information, posing a significant challenge to its generalization capability. Most baseline models experience severe performance degradation on the CSAR-HiQ dataset, whereas CDGN achieves a relatively low degradation rate (13.1% compared to PDBbind *core set*) while maintaining highest performance. Furthermore, in such a more challenging scenario, CDGN achieves a higher level of significance difference ($^{**}P = 9.9 \times 10^{-4}$ for CSAR-HiQ set compared to $^{*}P = 2.7 \times 10^{-2}$ for PDBbind *core set*) versus the best baseline. Note that in Table 3, we obtain similar conclusions that the significance differences in experimental results under the CSAR-HiQ grouping remain higher than that under the PDBbind grouping.

While the availability of 3D data for complex structure increases, the quantity of good-quality data within the *refined set* remains limited. Therefore, the model's capacity to leverage lower-quality data to enhance its performance underscores its generalizability, which shows extra vital dimension of generalization performance comparison. Therefore, we have conducted an additional experiment to assess the model's capacity for generalization, utilizing the *general set* from PDBbind. As shown in Fig. 5, we have compared CDGN with prominent competitive baseline models, employing two distinct training datasets. The outcomes demonstrate that CDGN consistently achieves substantially reduced prediction errors in both training scenarios. Moreover, our model demonstrates an average performance improvement of around 10% training on *general set*, solidifying its advantage over the baselines furtherly. Furthermore, under two indicators, CDGN achieves superior significance ($^{****}P = 5.3 \times 10^{-5}$ and $^{***}P = 4.5 \times 10^{-4}$) versus the second-best baseline in the lower-quality *general set*. As a result, CDGN exhibits greater generalizability to larger quantities of data with lower quality.

In summary, CDGN not only performs exceptionally well on the available high-quality data but also showcases its ability to leverage and improve performance using unseen and lower-quality data.

The superior generalization performance reflects CDGN's ability to extract rich biochemical information, making it suitable for real-world deployment scenarios.

4.3. Effectiveness of innovative components (RQ2)

In this subsection, we utilize GraphDTA (GCN) as an additional backbone to further assess the effectiveness of our innovative components. GraphDTA (GCN) leverages molecular graphs to directly capture the chemical bonds between atoms, representing drugs, and utilizes amino acid sequences to encode proteins. The representation vectors of the drug and protein are concatenated to model intramolecular interactions for predicting CPIs. In our experiment, we conduct ablation studies on each component of CDGN and GraphDTA by incorporating the intra-type channel, inter-type channel, and co-guided learning schema, respectively. We present the experimental results of each model with various configurations in Table 3. The *P*-values for all groups are much smaller than 0.05, indicating that adding or removing key components significantly affects the model performance. Below is a detailed analysis of each component.

4.3.1. Effectiveness of dual-channel mechanism

The dual-channel design in our model enhances performance by explicitly modeling both intramolecular and intermolecular interactions. As shown in Table 3, our dual-channel design (Model-1) improves the performance compared to GraphDTA. This improvement can be attributed to the fact that our design captures more comprehensive interaction features, including both intramolecular and intermolecular features, which are crucial for accurate and reliable CPI predictions. The results obtained from Model-3 and Model-4 demonstrate that the absence of either channel leads to a degradation in performance. This indicates that solely modeling either intramolecular or intermolecular interactions is insufficient for the model to fully learn the complex molecular interaction process. Therefore, our dual-channel design effectively incorporates both types of interactions, enabling the model to better capture the intricacies of molecular interactions and improve overall performance.

To provide a more intuitive analysis of the effectiveness of the dual-channel design in our model, we randomly select 50 complexes from the PDBbind *core set* and visualize the clustering distribution of the embedding vectors of the three types of edges used to model molecular interactions. We demonstrate this visualization before training (Fig. 6(a)) and after training (Fig. 6(b)) using t-distributed stochastic neighbor embedding (t-SNE) [43] in a 2D space. As shown in the Fig. 6(b), CDGN accurately identifies the three types of edges with the dual-channel design. Specifically, the embedding distribution of covalent bond edges, which represent intramolecular interactions between compounds and proteins, exhibits a similar pattern. On the other hand, the distribution of edge embeddings in the compound–protein bipartite graph, which models intermolecular interactions, shows a distinct pattern.

4.3.2. Effectiveness of co-guided learning schema

The co-guided learning schema improves performance by capturing the complex relationships between intramolecular and intermolecular interactions. To validate the effectiveness of this schema, we incorporate it into Model-1, resulting in Model-2. From Table 3, we can observe that the performance of GraphDTA is further improved with the inclusion of the co-guided learning schema. Furthermore, we replace the original co-guided learning schema in CDGN with a simple vector concatenation, represented as Model-5, whose performance decreases by 3.1% and 1.6% compared to CDGN in terms of RMSE on PDBbind and CSAR-HiQ datasets, respectively. This reduction in performance indicates that the absence of the co-guided learning mechanism prevents the model from effectively capturing the mutual influence and synergistic effects between the two types of interactions, leading to a decline in performance. These results further validate that our model successfully learns different edge representations to effectively model the two types of interactions separately. The dual-channel design enables proper differentiation and representation of both intramolecular and intermolecular interactions in the model. Besides, the co-guided learning schema enhances the model's ability to capture the intricate relationships between these interactions, resulting in improved performance.

To further demonstrate the model's ability to capture complex relationships of two types of interaction, we visualize the initial interaction vectors and the learned ones of the last MLP layer with and without the co-guided learning schema for 290 complexes in PDBBind *core set*. We obtain 256-dimensional representation vectors for each examined CPI and apply the Pearson correlation coefficient to any two vectors to gauge the similarity of each other. It is evident in Fig. 7(a) that, prior to model training, the similarity scores for most complex pairs' learned interactions are approximately 1.0. This suggests that disparate complexes exhibit nearly identical interaction patterns, and the untrained CDGN struggles to differentiate between them effectively. Further observation as shown in Fig. 7(b) reveals that the visualization of the similarity matrix presents certain levels of chaos after model without co-guided learning schema training, indicating that distinct complexes may possess varying interaction patterns, and the model exhibits a certain degree of capability to differentiate between these complexes based on the learned interaction patterns. However, after training the CDGN model, the similarity matrix of the complexes exhibits a higher level of chaos (as depicted in Fig. 7(c)). This suggests that our model has learned more nuanced interaction patterns, enabling it to better model intricate molecular interactions and, consequently, enhance its performance. This finding aligns with established domain knowledge that different compounds typically engage in distinct interactions with their respective targets.

4.4. Interpretability of CDGN (RQ3)

Despite achieving best results on two datasets, the interpretability of learned interaction patterns in the CDGN remains a worthwhile exploration due to the 'black-box' nature of deep learning models. To address this, we visualize the true interaction sites of the 2xnb complex as an example in Fig. 8. These atom–residue interaction sites are identified via Schrodinger software and visualized using PyMOL software. Additionally, to gain a deeper understanding of the contributions made by interactions between compound and protein atom pairs to the final prediction, we have conducted an analysis of the weighted weights for any atom–atom pair within the 2xnb. These weights are computed using Eq. (18). For a clearer representation, we have chosen to visualize atom–residue interaction weights instead of atom–atom's. We have employed a computational method for each atom–residue pair weight, which averages the weights of all potential atom–atom interactions within the atom–residue site. Similarly, we have computed certain interaction weight for one compound atom by averaging its all interactions with corresponding residue, and for each protein atom by averaging its interactions with each compound atom. Subsequently, we have selected the top 20 entities based on the average weight of compound atoms and protein residues, and we have presented the interaction weights between atoms and residues in Fig. 9(a).

In Fig. 9(a), the average atom–residue interaction weights indicate that certain residues (LEU-83, PHE-82, LYS-33, LEU-32, VAL-79, and HIS-84) and atoms (N-3, N-5, N-7, and O-28) have significantly stronger interaction weights compared to others. Interestingly, these residues and atoms align with the true interaction sites shown in Fig. 8, where residues LEU-83 and LYS-33, as well as compound atoms N-3, N-7, and O-28, are involved. Moreover, in Fig. 9(b), we visualize the main compound atom weights. We can notice that CDGN assigns greater interaction weights to the two benzene systems in comparison to the

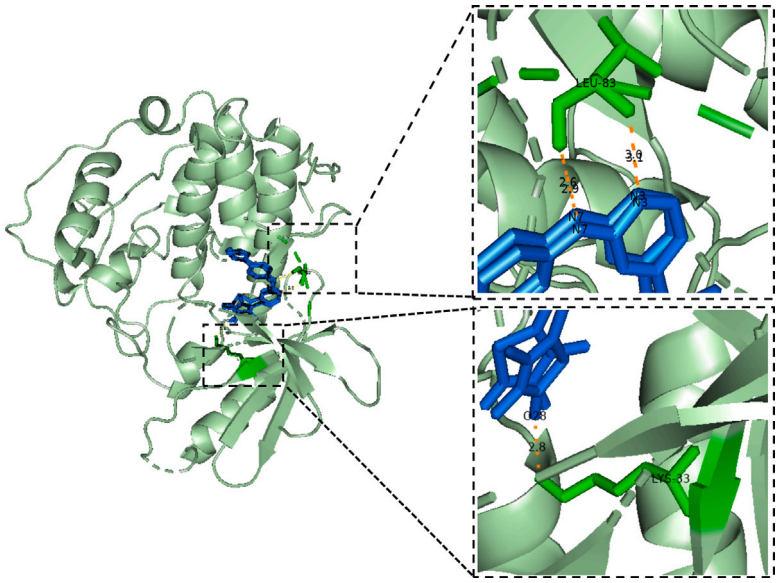


Fig. 8. Visualization of 3D structure of two real interaction sites between protein residues and compound atoms in 2xnb from PDBbind dataset. Dotted line in orange represents hydrogen bond interaction and the number is the corresponding bond length.

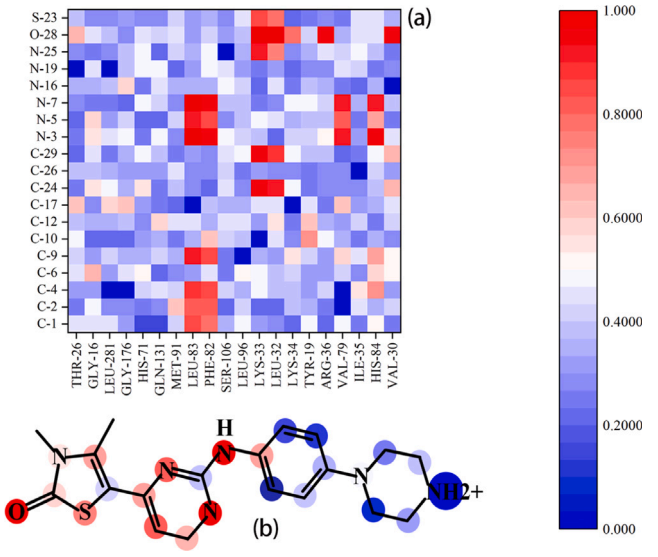


Fig. 9. (a): Heat map plot of interactions of 20 compound atoms and protein residues from 2xnb, the X axis is the pocket residue information, the Y axis is the compound atom symbol information. (b): Visualization of weights of the key atoms of compound molecular graph from 2xnb.

other elements within the compound molecule. This discovery aligns with the outcomes illustrated in Figs. 8 and 9(a).

The deviation of the predicted interaction sites from the true labels in this case could indeed be attributed to the scarcity of training data available to support the predictions. Despite this, the visualization results presented still demonstrate that the 'important' regions of CDGN predictions align with their corresponding true binding site labels to some extent. While the model may not always perfectly predict the exact interaction sites, the consistent alignment between the predicted interaction weights and the true binding sites, as shown in the visualizations, indicates that the CDGN model is able to capture meaningful and interpretable patterns in compound–protein interactions. This supports the notion that CDGN can effectively learn and leverage complex interaction patterns to make accurate predictions, even in cases with limited training data. Overall, the CDGN model demonstrates certain interpretability in binding sites, given the available data, further highlighting its efficacy in understanding and modeling CPIs.

4.5. Hyper-parameter study (RQ4)

In this subsection, we investigate the performance variation for CDGN w.r.t several necessary hyper-parameters by varying each parameter while keeping others fixed as default settings.

4.5.1. Cutoff distance r_θ

In different cutoff distance configurations, the compound–protein interaction graph exhibits varying scales, which subsequently affects the accuracy of experimental results and the time required for the experiment. We conduct an analysis on the effect of cutoff distance in constructing the interaction graph, varying r_θ from 1 Å to 10 Å. From Fig. 10(a), we can see that as r_θ increases, our model has access to more spatial information within the complex, which is favorable for the representation learning of complex and results in significant performance promotion when $r_\theta \leq 8$ Å. However, beyond this threshold, an excessively long cutoff distance introduces redundancies and degrades the performance. Further observations highlight that the performance improvement is particularly significant when $3 \leq r_\theta \leq 8$ Å. This finding aligns with the distribution of edge lengths mentioned in introduction, where non-bonding interactions tend to occur at distances greater than 2 Å. Similarly, as r_θ increases, more edges are taken into account in our graph, leading to longer propagation times for message passing in our model.

4.5.2. Number of intra-type channel layers k

Fig. 10(b) illustrates the impact of stacked convolution layers in the intramolecular module, ranging from 1 to 4 layers. It is observed that increasing the number of layers does not always lead to improved results. The model with only one layer has limited capability to capture high-order information in the complex. However, when using more than 3 layers, the model tends to overfit and the performance gradually deteriorates. Therefore, employing 2 convolution layers in the intra-type aggregating mechanism is sufficient to capture an adequate amount of intramolecular interaction information for subsequent processes.

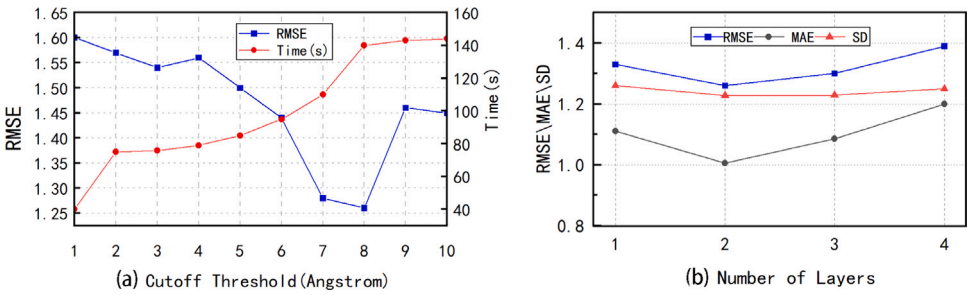


Fig. 10. Parameter analysis on PDBbind *core set*.

5. Conclusion

In our work, we aim to enhance the predictive accuracy of CPIs. To achieve this, we present a novel GNN-based model called CDGN. CDGN incorporates a dual-channel aggregating mechanism that captures both intramolecular and intermolecular interaction information. The intra-type channel captures intramolecular interactions, while the inter-type channel focuses on intermolecular interactions. To further enhance the learning process, we introduce a co-guided learning schema that facilitates the modeling of complex correlations between different interactions. This schema allows the intramolecular and intermolecular interactions to mutually reinforce and improve the learning of each other. Comprehensive experiments on two datasets demonstrate the superiority of CDGN over existing state-of-the-art methods in terms of predicting CPIs.

In summary, CDGN has achieved the following important contributions in predicting CPIs: (1) Guided by biochemical prior knowledge and combined with detailed statistical analysis of datasets and physicochemical structures of biological samples, we have clarified the importance of co-modeling intramolecular and intermolecular interactions to provide comprehensive information for CPIs. (2) To realize insights into intra-molecular and inter-molecular interaction mechanisms, we adopt a dual-channel modeling approach, enabling the model to capture semantic information in the covalent environment in the intramolecular channel and perceive noncovalent interaction information related to 3D distance in the intermolecular channel. Furthermore, we employ a co-guided mechanism to capture the complex correlations between the two types of interaction information, thus synergistically assisting CPI predictions. (3) We conducted extensive experiments on two public benchmark datasets, validating our model's performance and generalization ability relative to baseline methods in challenging experimental scenarios (unseen data and low-quality data). Additionally, we conducted statistical analysis, case visualization, and detailed studies on key components and hyperparameters based on experimental results.

CDGN takes structured compound–protein 3D complex graphs as input, which brings advantages and limitations to our method. On one hand, the high-information content of structured input allows our model to preliminarily locate the binding pocket region of the protein, which helps the model infer the detailed binding mechanism between proteins and compounds, and speeds up convergence by ignoring some non-binding regions. On the other hand, our model requires the co-structure of proteins and compounds as input, but the limited number of resolved complex structures may still restrict the application range of current structure-based models.

In our future works, to address the problem of limited high-quality 3D structural data, we will further adopt unsupervised or semi-supervised approaches to extract valuable information from structure-free data (such as molecular 2D graphs and protein sequences), and combine it with other tasks such as binding site prediction to synergistically improve CPI predictions.

CRediT authorship contribution statement

Zheyu Wu: Conceptualization, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing. **Huifang Ma:** Funding acquisition, Supervision. **Bin Deng:** Data curation, Methodology, Validation. **Zhixin Li:** Funding acquisition. **Liang Chang:** Funding acquisition.

Declaration of competing interest

We would like to submit the manuscript entitled 'Co-guided Dual-channel Graph Neural Networks for the Prediction of Compound–Protein Interaction', which we wish to be considered for publication in Applied Soft Computing Journal. No conflict of interest exists in the submission of this manuscript, and manuscript is approved by all authors for publication.

Data availability

The datasets used in this paper are all public.

Acknowledgments

This study is supported by the Gansu Province Key Foundation Project (No. 24JRRA123), Industrial Support Project of Gansu Colleges, China (No. 2022CYZC11), National Natural Science Foundation of China (61762078, 62276073 and U22A2099) and Graduate Research Funding of Northwest Normal University (2023KYZZ—S121).

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